Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-7 | Date: 16-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**A**	3	**C**	4	**C**	5	**A**
6	**B**	7	**C**	8	**C**	9	**A**	10	**B**
11	**D**	12	**C**	13	**C**	14	**B**	15	**C**
16	**D**	17	**B**	18	**B**	19	**D**	20	**B**
21	**25**	22	**2**	23	**77**	24	**8**	25	**1**

CHEMISTRY

26	**D**	27	**B**	28	**B**	29	**B**	30	**C**
31	**C**	32	**B**	33	**C**	34	**D**	35	**C**
36	**B**	37	**A**	38	**D**	39	**A**	40	**A**
41	**C**	42	**C**	43	**B**	44	**B**	45	**C**
46	**8**	47	**234**	48	**25**	49	**2**	50	**0**

MATHEMATICS

51	**C**	52	**A**	53	**B**	54	**B**	55	**A**
56	**B**	57	**C**	58	**C**	59	**D**	60	**B**
61	**D**	62	**B**	63	**B**	64	**A**	65	**B**
66	**A**	67	**B**	68	**B**	69	**C**	70	**A**
71	**320**	72	**26**	73	**1**	74	**16**	75	**18915**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. $2\sqrt{\frac{L}{g}}$

$$v = \frac{dx}{dt} = \sqrt{xg}$$

$$\int_0^T dt = \frac{1}{\sqrt{g}} \int_0^L x^{-1/2} dx$$

$$T = \frac{1}{\sqrt{g}} \left[\frac{x^{1/2}}{1/2} \right]_0^L$$

$$T = \frac{1}{\sqrt{g}} [2\sqrt{x}]_0^L$$

$$T = 2\sqrt{\frac{L}{g}}$$

2. To find the heat capacity, we use the formula:

$$\text{Heat Capacity } (C) = \frac{\text{Total Heat Energy } (Q)}{\text{Temperature Change } (\Delta T)}$$

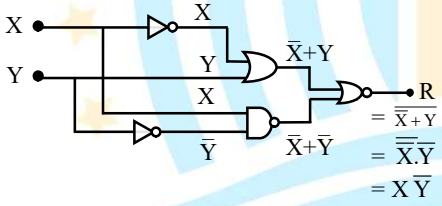
1. Total Heat (Q): $1000\ \text{W} \times 120\ \text{s} = 120{,}000\ \text{J}$
2. Temperature Change (ΔT): $31^\circ\text{C} - 25^\circ\text{C} = 6^\circ\text{C}$

Calculation:

$$C = \frac{120{,}000\ \text{J}}{6^\circ\text{C}} = 20{,}000\ \text{J}/^\circ\text{C} = 2 \times 10^4\ \text{J}/^\circ\text{C}$$

3. $(3L)\alpha_{eff}\Delta\theta = L\alpha\,\Delta\theta + 2L(2\alpha)(\Delta\theta)$

$\therefore\ \alpha_{eff} = \frac{5}{3}\alpha$

4. 

* Output R = $X\ \bar{Y}$
* If R = 1 it is possible when X = 1 & Y= 0

5. • Induced EMF (ϵ):

$$\epsilon = Blv = 0.15 \times 0.5 \times 2 = 0.15\ \text{V}$$

• Induced Current (I):

$$I = \frac{\epsilon}{R} = \frac{0.15}{3} = 0.05\ A$$

• Magnetic Force (F):

To maintain constant velocity, the external force must equal the magnetic force:

$$F = BIl = 0.15 \times 0.05 \times 0.5$$

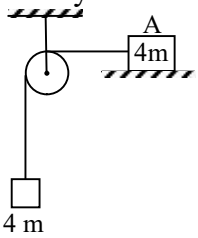
$$F = 0.00375\ \text{N} = 3.75 \times 10^{-3}\ \text{N}$$

6. $KE_{max} = h(\nu - \nu_0)$

$$\frac{KE'_{max}}{KE_{max}} = \frac{h(2\nu - \nu_0)}{h(\nu - \nu_0)} > 2$$

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7. The system can be redraw as



$\therefore a = \frac{4mg}{8m} = \frac{g}{2} = 5\ m/s^2$

$\therefore v = u + at = 2 \times 5 = 10\ m/s$

8. System: The gas expands into a vacuum (evacuated chamber). This is a case of Free Expansion.

1. Work (W): Since the external pressure is zero ($P_{ext} = 0$), the work done by the gas is $W = 0$.
2. Heat (Q) : The cylinder is non-conducting, so no heat enters or leaves: $Q = 0$.
3. Internal Energy (ΔU): From the First Law ($\Delta U = Q - W$), we get:

$\Delta U = 0 - 0 = 0$

For an ideal gas, internal energy depends only on temperature. If $\Delta U = 0$, then the temperature change $\Delta T = 0$.

- Initial Temperature: 300 K

Final Temperature: 300 K

9. At $t = 0$:

- Capacitor acts as a short circuit and Inductor acts as an open circuit.
- Total resistance: $R_1 + R_2 = 5\Omega + 5\Omega = 10\Omega$.
- Current $I(0) = \frac{20\ V}{10\Omega} = 2\ A$.

At $t = \infty$:

- Capacitor acts as an open circuit and Inductor acts as a short circuit.
- Total resistance: $R_1 + R_3 = 5\Omega + 5\Omega = 10\Omega$.
- Current $I(\infty) = \frac{20\ V}{10\Omega} = 2\ A$.

Ratio:

$\frac{I(0)}{I(\infty)} = \frac{2\ A}{2\ A} = 1$

10. Given the circuit is a series circuit:

1. Total Resistance (R_{total}):

$R_{total} = \frac{V_1}{A_1} = \frac{26\ V}{2\ A} = 13\Omega$

2. Calculate x :

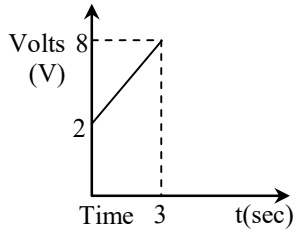
Since resistors are in series:

$x + 3\Omega + 6\Omega = 13\Omega$

$x + 9 = 13$

$x = 4\Omega$

11.



$I = \frac{dq}{dt}$

$q = it + a$

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$V = \frac{q}{c}$

$V = \frac{it+a}{c}$

$\therefore$ V is proportional to time.

12. Given $q_1 + q_2 = Q$ and surface densities $\sigma_1 = \sigma_2$:

$$\frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2} \Rightarrow \frac{q_1}{r^2} = \frac{q_2}{R^2} = \frac{q_1+q_2}{r^2+R^2} = \frac{Q}{r^2+R^2}$$

Thus, $q_1 = \frac{Qr^2}{r^2+R^2}$ and $q_2 = \frac{QR^2}{r^2+R^2}$.

The potential at the common centre is:

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{R}\right)$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{Qr}{r^2+R^2} + \frac{QR}{r^2+R^2}\right)$$

$$V = \frac{Q(r+R)}{4\pi\epsilon_0(r^2+R^2)}$$

13. To hear beats, the observer must perceive two different frequencies from sources S_1 and S_2.

Using the Doppler effect formula $f_{obs} = f\left(\frac{v \pm v_o}{v}\right)$: :

1. Positions A and C (Outside)

- At A , the observer moves toward both S_1 and S_2. Thus, $f_1 = f_2$ and no beats are heard.
- At C , the observer moves away from both S_1 and S_2. Thus, $f_1 = f_2$ and no beats are heard.

2. Position B (Between)

- The observer moves away from S_1 (f_1 decreases) and toward S_2 (f_2 increases). Since $f_1 \neq f_2$, beats will be heard.

14. Apparent weight (w_a)

= actual weight (w) – upthrust (F)

Here, $F = V\rho_w g$ (ρ_w = density of water)

i.e., $F_{0^\circ C} = V_{0^\circ C}(\rho_w)_{0^\circ C} g$

and $F_{50^\circ C} = V_{50^\circ C}(\rho_w)_{50^\circ C} g$

$$\therefore \frac{F_{50^\circ C}}{F_{0^\circ C}} = \frac{V_{50^\circ C}}{V_{0^\circ C}} \cdot \frac{(\rho_w)_{50^\circ C}}{(\rho_w)_{0^\circ C}} = \frac{(1+\gamma_m \Delta\theta)}{(1+\gamma_w \Delta\theta)} \quad (\Delta\theta = 50^\circ C)$$

Given that $\gamma_m < \gamma_w \therefore F_{50^\circ C} < F_{0^\circ C}$

or apparent weight at 50° C will be more.

15. At 'B' shape of pulse gets inverted. The pulse will become erect after reflection at 'A'.

16. By the principle of calorimetry, Heat lost = Heat gained. Let T_f be the final temperature.

$$m \cdot C \cdot (2T - T_f) = \left(\frac{m}{2}\right) \cdot (2C) \cdot (T_f - T)$$

$$m \cdot C \cdot (2T - T_f) = m \cdot C \cdot (T_f - T)$$

$$2T - T_f = T_f - T$$

$$3T = 2T_f$$

$$T_f = \frac{3}{2}T$$

17. Time period of both A and B is $T = 2\pi\sqrt{\frac{\ell}{g}}$

After first collision, B acquires amplitude of A and after second collision it acquires its own amplitude in this process time taken is

$$= \frac{T}{4} + \frac{T}{4} + \frac{T}{4} + \frac{T}{4} = T = 2\pi\sqrt{\frac{\ell}{g}}$$

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18. $\text{P.E.} = \frac{Y}{2}(\text{strain})^2(AL) = \text{K.E.} = \frac{1}{2}mv^2$

$v = \text{strain}\sqrt{\frac{Y}{m}AL}$

$= \frac{2}{10}\sqrt{\frac{5\times10^8}{5\times10^{-3}}\times10^{-6}\times0.1}$

$= 20$ m/s

19. Angular velocity of P with respect to Q is given by:

$\omega = \frac{v_{\perp\,rel}}{r}$

1. Perpendicular components of velocities:

- $v_{P\perp} = 8\sin 30° = 4\ m/s$ (downwards)
- $v_{Q\perp} = 6\sin 30° = 3\ m/s$ (upwards)

2. Relative perpendicular velocity:

$v_{\perp rel} = 4 + 3 = 7\ m/s$

3. Angular velocity:

$\omega = \frac{7}{10} = 0.7 rad/sec$

20. The Earth revolves around the Sun once a year and its orbit has a mean radius 150×10^6 km. Hence its average orbital speed is

$$v = \frac{2\pi\times150\times10^9}{365\times24\times3600} = 3 \times 10^4 \text{ m s}^{-1}$$

As the Earth has mass 6×10^{24} kg, its average momentum is

$$p = mv = 1.8 \times 10^{29} \text{ kg m s}^{-1}$$

and so its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.63\times10^{-34}}{1.8\times10^{29}} = 3.7 \times 10^{-63} \text{ m},$$

If the Sun can be considered stationary.

21. The magnification required is:

$M = \frac{D_2}{D_1} = \frac{15\ mm}{3\ mm} = 5$

Since $M = \frac{f_2}{f_1}$, we can substitute our values:

$5 = \frac{125\ mm}{f_1}$

22. Formula for root mean square velocity (v_{rms}) :

$v_{rms} = \sqrt{\frac{3P}{\rho}}$

Calculation:

$v_{rms} = \sqrt{\frac{3\times2\times10^6}{1.5}}$

$v_{rms} = \sqrt{\frac{6\times10^6}{1.5}}$

$v_{rms} = \sqrt{4\times10^6}$

$v_{rms} = 2\times10^3\ m/s$

23. Energy per fission

$E_0 = 200\times 10^6 \times 1.6 \times 10^{-19}$ Joule

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$E_0 = 3.2 \times 10^{-11}$ Joule

$P = 4$ MW $= 4 \times 10^6$ Joule/s

In time t (= 1 year) energy delivered = Pt

Number of fission required = $\frac{Pt}{E_0}$

$= \frac{4\times10^6\times365\times24\times3600}{3.2\times10^{-11}}$

$= 3.942 \times 10^{24}$

Mass of uranium required

$= \frac{3.942\times10^{24}\times235}{6.02\times10^{23}}$ gm

$= 1538.8$ gm

$\simeq 1539$ gm

24. $P = k\,\omega^a \rho^b \ell^c$

$\Rightarrow \quad ML^2T^{-3} = (T^{-1})^a (ML^{-3})^b L^c$

$\Rightarrow \quad a = 3 \quad \therefore P \propto \omega^3$

25. Let $\lambda_A = \lambda$ and $\lambda_B = 2\lambda$

Initially rate of disintegration of A is λN_0 and that of B is $2\lambda N_0$.

After one half-life of A, rate of disintegration of A will becomes $\frac{\lambda N_0}{2}$ and that of B would also be $\frac{\lambda N_0}{2}$ so after one half-life of A or two half-lives of B. $\left(\frac{-dN}{dt}\right)_A = \left(\frac{-dN}{dt}\right)_B$

$\therefore n = 1$

CHEMISTRY

26. Conceptual

27. The oxide of chromium is Cr_2O_3. Each chromium has $12e^-$ and each oxygen has $6e^-$. Total electrons with $l = 1$ are 42

28. Dipole moments of HCHO, CH_3CHO and CH_3COCH_3 are 2.3, 2.7 and 2.9D respectively.

29. For complex A, CSFE = $-1.2\Delta_0$

For complex B, CSFE = $-0.8\Delta_t$

Ratio of their CSFE = $\frac{-1.2\Delta_0}{-0.8\Delta_t} = \frac{-1.2\Delta_0}{-0.8(\frac{4}{9}\Delta_0)} = \frac{27}{8}$

30. Conceptual

31. $\ln\left[\frac{\left(t_{1/2}\right)_1}{\left(t_{1/2}\right)_2}\right] = \frac{Ea}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

$\ln\left[\frac{3}{\left(t_{1/2}\right)_2}\right] = \frac{840R}{R}\left[\frac{1}{300} - \frac{1}{400}\right]$

$\left(t_{1/2}\right)_2 = 1.5$ min

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32. 6 moles of solute present in 1000 ml solution

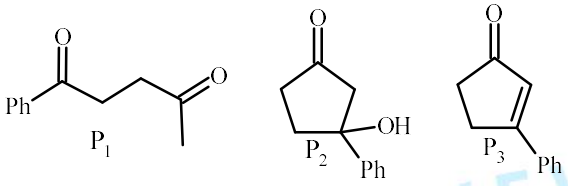
$\because$ no change in volume

6 moles of solute is dissolved in 1000 ml or 1000d g of solvent

$$m = \frac{n}{W(g)} \times 1000 \Rightarrow 7.5 = \frac{6}{1000d} \times 1000$$

$\therefore$ Density $d = \frac{6}{7.5} = 0.8\ \text{g/cm}^3$

33.



34. $$\log\frac{P_2}{P_1} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\frac{10}{1} = \frac{921.2}{2.303\times 2}\left[\frac{1}{100} - \frac{1}{T_2}\right] \quad \therefore \quad T_2 = 200\text{K}$$

35. In solution b and d we will get equimolar mixture of weak acid and its conjugate base, hence pH =pKa

36. Number of millimoles of ketose = no.of millimoles of ketoxime

i.e. Millimoles of ketoxime = $\frac{0.45}{150} \times 1000 = 3$

$\therefore$ Millimoles of aldoxime = millimoles of aldose = 2

Thus, the weight of aldose in the mixture = $\frac{2\times 150}{1000} g$

37. Conceptual
38. Conceptual
39. HOMO molecular orbital of more substituted alkene is more nucleophilic in nature
40. Conceptual
41. Conceptual
42. Conceptual
43. Conceptual
44. Conceptual
45. Conceptual
46. $\Delta H_{dissolution} = \Delta H_{(ionisation)} + \Delta H_{(hydration)}$

$= 780 - 775 = 5\ \text{kJ mol}^{-1}$

$\Delta G_{dissolution} = \Delta H_{dissolution} - T\Delta S_{dissolution}$

$= 5000 - 298 \times 45 = -8410\ \text{J}$

$= 8.41\ \text{kJ mol}^{-1}$

47. Conceptual
48. Conceptual
49. Conceptual
50. $$T_A = \frac{2P_0V_0}{R}; \quad T_B = \frac{2P_0V_0}{R}$$

$T_A - T_B = 0$

$\therefore \quad \Delta U = 0$

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MATHS

52. $\tan\theta + \sqrt{5}\tan\theta \cdot \tan 2\theta = \sqrt{5} - \tan 2\theta$

$\tan\theta + \tan 2\theta = \sqrt{5}(1 - \tan\theta\tan 2\theta)$

$$\frac{\tan\theta + \tan 2\theta}{1 - \tan\theta\tan 2\theta} = \sqrt{5}$$

$\tan 3\theta = \sqrt{5} = \tan\alpha$, When $\alpha = \tan^{-1}\sqrt{5}$

$3\theta = n\pi + \alpha \Rightarrow \theta = \frac{n\pi}{3} + \frac{\alpha}{3}$

for $n = 0, \theta = \frac{\alpha}{3} \in S$; for $n = 1, \theta = \frac{\pi}{3} + \frac{\alpha}{3} \in S$

for $n = -1, \theta = -\frac{\pi}{3} + \frac{\alpha}{3} \in S$

for $n = -2, \theta = -\frac{2\pi}{3} + \frac{\alpha}{3} \in S$

for $n = -3, \theta = -\pi + \frac{\alpha}{3} \in S$

53. for $p > 0, \bar{v}_2$ is obtained by rotating $\bar{v}_1$ by an angle θ,

$\therefore |\bar{v}_1| = |\bar{v}_2|$

$\Rightarrow 3p^2 + 1 = 4 + (p+1)^2$

$\Rightarrow p = 2, (\because p \neq -1 \text{ as } p > 0).$

Now, angle b/w. $\bar{v}_1 \& \bar{v}_2$ is given by, $\cos\theta = \frac{\bar{v}_1 \cdot \bar{v}_2}{|\bar{v}_1||\bar{v}_2|}$.

$$\cos\theta = \frac{4\sqrt{3}+3}{13}$$

$$\tan\theta = \frac{\sqrt{112 - 24\sqrt{3}}}{4\sqrt{3}+3}$$

$$= \frac{6\sqrt{3}-2}{4\sqrt{3}+3} = \frac{\alpha\sqrt{3}-2}{4\sqrt{3}+3}$$

56. $|A| = 54 - 18a$,

If $|A| = 0$, then $a = 3$ and system have infinitely many solutions.

If $|A| \neq 0$ then $a \neq 3$, and system has unique solution.

$$\text{adj}A = \begin{bmatrix} 19 & -38 & 0 \\ 4 & -8 & 0 \\ -9 & 18 & 0 \end{bmatrix}$$

for infinitely many sol: $(\text{adj}A)B = 0$

$$\Rightarrow \begin{bmatrix} 19 & -38 & 0 \\ 4 & -8 & 0 \\ -9 & 18 & 0 \end{bmatrix} \begin{bmatrix} b \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow b = 8$

57.

$$1 + \log_{(x+1)}(2x+5) + \frac{2}{\log_{(x+1)}(2x+5)} = 4$$

$\log_{(x+1)}(2x+5) = t, t^2 - 3t + 2 = 0 \Rightarrow t = 1, 2.$

$t = 1 \Rightarrow x = -4 (x > 0)$

$t = 2 \Rightarrow x = 2$

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58. $x^2+y^2+6x+8y+16=0 \to c_1(-3,-4), r_1=3$

$x^2+y^2+2(3-\sqrt{3})x+2(4-\sqrt{6})y=k+6\sqrt{3}+8\sqrt{6} \to c_2(-3+\sqrt{3}-4+\sqrt{6})$

$r_2=\sqrt{34+k}.$

circles touch internally. $c_1c_2=|r_2-r_1|$

$k=2, r_2=6.$

Distance b/w, the centre of circle in 3 and $r_1=3$,

So, $(-3,-4)$ is the mid point of $(-3+\sqrt{3},-4+\sqrt{6})$ and (α,β)

$\alpha=-3-\sqrt{3}.$

$\beta=-4-\sqrt{6},$

$(\alpha+\sqrt{3})^2+(\beta+\sqrt{6})^2=25.$

59.

$$\left(-\frac{\mu}{2}\right)^2=\left(\frac{b}{a}\right)^2\left(\frac{\lambda}{2}\right)^2-b^2$$

$$\Rightarrow\frac{\mu^2}{b^2}=\frac{\lambda^2}{a^2}-4$$

$$\left(\frac{\lambda}{a}\right)^2-\left(\frac{\mu}{b}\right)^2=4.$$

60.

$(2\bar{a}+3\bar{b})\cdot\bar{c}=1670, \quad (\bar{a}+\bar{b})\times\bar{c}=\bar{c}\times(-2\bar{a}+3\bar{b})$

$(2\bar{a}+3\bar{b})\cdot\lambda(4\bar{b}-\bar{a})=1670, \ \bar{c}=\lambda(4\bar{b}-\bar{a})$

$\lambda=1$

$\bar{c}=4\bar{b}-\bar{a}$

61. Let $P(E_1)=P_1; P(E_2)=P_2; P(E_3)=P_3$

$P(E_1\cap\bar{E}_2\cap\bar{E}_3)=\alpha=P_1(1-P_2)(1-P_3) \ -(i)$

$P(\bar{E}_1\cap E_2\cap\bar{E}_3)=\beta=(1-P_1)P_2(1-P_3) \ -(ii)$

$P(\bar{E}_1\cap\bar{E}_2\cap E_3)=\gamma=(1-P_1)(1-P_2)P_3 \ -(iii)$

and $P(\bar{E}_1\cap\bar{E}_2\cap\bar{E}_3)=p=(1-P_1)(1-P_2)(1-P_3) \ -(iv)$

Now, $(\alpha-2\beta)p=\alpha\beta$

$\Rightarrow[P_1(1-P_2)(1-P_3)-2(1-P_1)P_2(1-P_3)]p$

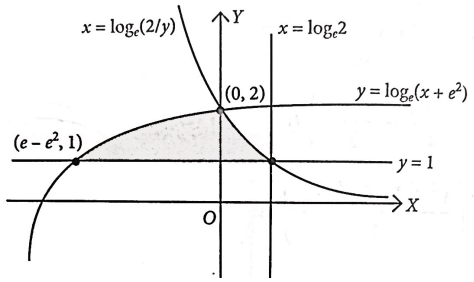
$=P_1P_2(1-P_1)(1-P_2)(1-P_3)^2$

$\Rightarrow P_1(1-P_2)-2(1-P_1)P_2=P_1P_2\Rightarrow P_1=2P_2 \ -(v)$

Also, $(\beta-3\gamma)p=2\beta\gamma\Rightarrow P_2=3P_3$ - (vi)

Hence, $P_1=2\times3P_3 \ \therefore\frac{P_1}{P_3}=6$

62.



For More Join: @IITJEE_Advanced

We have, $y = \log_e(x+e^2), x = \log_e(2/y)$,

$x = \log_e 2, y = 1$

Now, $x = \log_e \frac{2}{y} \Rightarrow e^x = \frac{2}{y} \Rightarrow y = 2e^{-x}$

$$\text{Required area } = \int_{e-e^2}^{0} [\log_e(x+e^2) - 1]dx + \int_{0}^{\log_e^2} (2e^{-x} - 1)dx$$

$$= [x \cdot \log_e(x+e^2)]_{e-e^2}^{0} - \int_{e-e^2}^{0} \frac{x}{x+e^2}dx - [x]_{e-e^2}^{0}$$

$$+ \left[\frac{2 \cdot e^{-x}}{-1} - x\right]_0^{\log_e 2}$$

$$= 0 - (e-e^2) \cdot 1 - \int_{e-e^2}^{0} \left(\frac{x+e^2}{x+e^2} - \frac{e^2}{x+e^2}\right)dx - [0-(e-e^2)]$$

$$+ \left[\frac{2 \cdot 2^{-1}}{-1} - \log_e 2 + 2\right]$$

$$= e^2 - e - [x - e^2\log_e(x+e^2)]_{e-e^2}^{0} + e - e^2 - 1 - \log_e 2 + 2$$

$$= -[0 - 2e^2 - (e-e^2) + e^2 \cdot 1] + 1 - \log_e 2$$

$$= 2e^2 + e - e^2 - e^2 + 1 - \log_e 2 = (e + 1 - \log_e 2) \text{ sq. units}$$

63. Let $y = \frac{4}{\sin x} + \frac{1}{1-\sin x} = \frac{4-3\sin x}{\sin x(1-\sin x)}$

Let $\sin x = t$, where $t \in (0,1) \therefore y = \frac{4-3t}{t-t^2}$

Now, $\frac{dy}{dt} = \frac{(t-t^2)(-3)-(4-3t)(1-2t)}{(t-t^2)^2} = 0$

$$\Rightarrow 3t^2 - 3t - (4 - 11t + 6t^2) = 0$$

$$\Rightarrow 3t^2 - 8t + 4 = 0 \Rightarrow (3t-2)(t-2) = 0$$

$$\Rightarrow t = \frac{2}{3}\{t \in (0,1)\} \Rightarrow \sin x = \frac{2}{3}$$

$$\therefore y_{\min} = \frac{4}{2/3} + \frac{1}{1-2/3} = 9$$

$[\because y_{\max} \to \infty]$

$\Rightarrow \alpha_{\min} = 9$

64.

$$\text{Given, } f(x) = \int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}$$

$$x = \frac{1}{t} \Rightarrow dx = \frac{-1}{t^2}dt$$

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$$\Rightarrow f(x)=\int \frac{-\frac{1}{t^2}dt}{\frac{\left(3t^2+4\right)}{t^2}\frac{\sqrt{4t^2-3}}{t}}$$

$$=-\int \frac{-tdt}{(3t^2+4)\sqrt{4t^2-3}} \text{ Put } 4t^2-3=z^2\Rightarrow tdt=\frac{z}{4}dz$$

$$=-\frac{1}{4}\int \frac{zdz}{\left(3\left(\frac{z^2+3}{4}\right)+4\right)z}=\int \frac{-dz}{3z^2+25}=-\frac{1}{3}\int \frac{dz}{z^2+\left(\frac{5}{\sqrt{3}}\right)^2}$$

$$=\int -\frac{1}{3}\frac{\sqrt{3}}{5}\tan^{-1}\left(\frac{\sqrt{3}z}{5}\right)+C=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{5}\sqrt{4t^2-3}\right)+C$$

$$f(x)=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{5}\sqrt{\frac{4-3x^2}{x^2}}\right)+C$$

$$\because f(0)=0\therefore C=\frac{\pi}{10\sqrt{3}}$$

$$\text{Now, } f(1)=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)+\frac{\pi}{10\sqrt{3}}$$

$$\Rightarrow f(1)=\frac{1}{5\sqrt{3}}\cot^{-1}\left(\frac{\sqrt{3}}{5}\right)=\frac{1}{5\sqrt{3}}\tan^{-1}\frac{5}{\sqrt{3}}$$

$$\text{So, } \alpha=5:\beta=\sqrt{3}\therefore \alpha^2+\beta^2=28$$

65. We have, $\sum_{n=0}^{\infty}\frac{n^3(2n!)+(2n-1)(n!)}{(n!)((2n)!)}=ae+\frac{b}{e}+c$

$$\Rightarrow \sum_{n=0}^{\infty}\frac{1}{(n-3)!}+\sum_{n=0}^{\infty}\frac{3}{(n-2)!}+\sum_{n=0}^{\infty}\frac{1}{(n-1)!}$$

$$+\sum_{n=0}^{\infty}\frac{1}{(2n-1)!}-\sum\frac{1}{(2n)!}=ae+\frac{b}{e}+c$$

$$\Rightarrow e+3e+e+\frac{1}{2}\left(e-\frac{1}{e}\right)-\frac{1}{2}\left(e+\frac{1}{e}\right)=ae+\frac{b}{e}+c$$

$$\Rightarrow 5e-\frac{1}{e}=ae+\frac{b}{e}+c$$

Comparing, we get, $a=5,b=-1,c=0,a^2-b+c=25+1=26.$

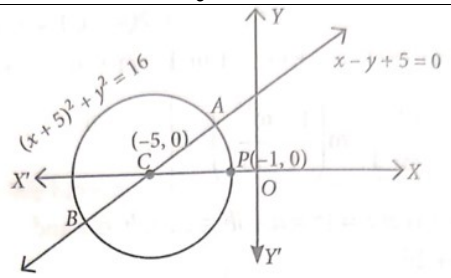
67. We have, $|z+5|\le 4$

$$\Rightarrow |x+iy+5|\le 4\Rightarrow \sqrt{(x+5)^2+y^2}\le 4$$

$$\Rightarrow (x+5)^2+y^2\le 16$$

Also, $z(1+i)+\bar{z}(1-i)\ge -10\Rightarrow (z+\bar{z})+i(z-\bar{z})\ge -10$

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$\therefore |z+1|^2 = |z-(-1)|^2$

Let $P(-1,0)$

$\therefore |z+1|^2_{max} = PB^2$ (where B is in 3rd quadrant)

For point of intersection, solving, $(x+5)^2+y^2=16$ and $x-y+5=0$, we get

$A(2\sqrt{2}-5, 2\sqrt{2})$ and $B(-2\sqrt{2}-5, -2\sqrt{2})$

Now, $PB^2 = (2\sqrt{2}+4)^2 + (2\sqrt{2})^2$

$\Rightarrow |z+1|^2 = 8+16+16\sqrt{2}+8$

$\Rightarrow \alpha+\beta\sqrt{2} = 32+16\sqrt{2}$

$\Rightarrow \alpha = 32$ and $\beta = 16$.

Hence, $\alpha+\beta = 48$

68.

Age	No. of Students	C.F.
15	5	5
16	8	13
17	5	18
18	12	30
19	x	$30+x$
20	y	$30+x+y$

$30+x+y=40$ (given)

$\Rightarrow x+y=10$ (i)

Median of the given data $= 18$

Mean deviation about median $= \frac{\Sigma f_i |x_i-18|}{\Sigma f_i}$

$\Sigma f_i|x_i-18| = 5\times3+8\times2+5\times1+12\times0+x\times1+y\times2$

$= 15+16+5+x+2y = 36+x+2y$

$\Rightarrow \frac{36+x+2y}{40} = 1.25$ (given)

$\Rightarrow x+2y=14$ (ii)

On solving (i) and (ii), we get

$x=6, y=4$

Now, $4x+5y = 4\times6+5\times4 = 24+20 = 44$

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69. Equations of parabola,

$$\sqrt{\left(x+\frac{1}{2}\right)^2+(y-0)^2}=\left|y+\frac{1}{2}\right| \Rightarrow x^2+\frac{1}{4}+x+y^2=y^2+\frac{1}{4}+y$$

$$\Rightarrow x^2+x=y=f(x) \text{ (let)}$$

Two Dimensional Geometry

Now, $\tan^{-1}\sqrt{f(x)}+\sin^{-1}\sqrt{f(x)+1}=\frac{\pi}{2}$

$$\cos^{-1}\frac{1}{\sqrt{1+f(x)}}+\sin^{-1}\sqrt{f(x)+1}=\frac{\pi}{2}$$

We know, $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$,

$$\Rightarrow \frac{1}{\sqrt{1+f(x)}}=\sqrt{f(x)+1} \Rightarrow f(x)+1=1 \Rightarrow f(x)=0$$

$$x^2+x=0 \Rightarrow x(x+1)=0 \Rightarrow x=0,-1 \Rightarrow S=\{-1,0\}$$

70. Given, $x^2\sin\theta-x(\sin\theta\cos\theta+1)+\cos\theta=0$

$$D=(1+\sin\theta\cos\theta)^2-4\sin\theta\cos\theta=(1-\sin\theta\cos\theta)^2$$

$$\therefore x=\frac{1+\sin\theta\cos\theta\pm(1-\sin\theta\cos\theta)}{2\sin\theta}$$

$\Rightarrow \alpha=\cos\theta$ and $\beta=\text{cosec}\theta$

Now, $\sum_{n=0}^{\infty}\left(\alpha^n+\left(-\frac{1}{\beta}\right)^n\right)=\sum_{n=0}^{\infty}(\cos\theta)^n+\sum_{n=0}^{\infty}(-\sin\theta)^n=\frac{1}{1-\cos\theta}+\frac{1}{1+\sin\theta}$

71. Given differential equation is

$$(1-x^2)dy=\left(xy+(x^3+2)\sqrt{1-x^2}\right)dx,-1<x<1$$

$$\frac{dy}{dx}=\frac{xy+(x^3+2)\sqrt{1-x^2}}{1-x^2} \Rightarrow \frac{dy}{dx}-\left(\frac{x}{1-x^2}\right)y=\frac{x^3+2}{\sqrt{1-x^2}}$$

$$\text{I.F.}=e^{\int\frac{-x}{1-x^2}dx}=\sqrt{1-x^2}$$

$\therefore$ The solution of the given D.E. is

$$y\cdot\sqrt{1-x^2}=\int(2+x^3)dx+c \Rightarrow y\sqrt{1-x^2}=2x+\frac{x^4}{4}+c-(i)$$

When $x=0,y=0 \therefore 0=0+0+c \Rightarrow c=0$

From (i) $y\sqrt{1-x^2}=2x+\frac{x^4}{4}$

Now, $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(2x+\frac{x^4}{4}\right)dx=\left[x^2+\frac{x^5}{20}\right]_{-\frac{1}{2}}^{\frac{1}{2}}=\frac{1}{320}$

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72. Equation of the plane passing through the line of intersection of given plane is

$(2x+y-z-3)+\lambda(5x-3y+4z+9)=0$

$\Rightarrow x(2+5\lambda)+y(1-3\lambda)+z(4\lambda-1)+9\lambda-3=0-(i)$

The plane is parallel to the line

$\frac{x+2}{2}=\frac{3-y}{-4}=\frac{z-7}{5}$ i.e., $\frac{x+2}{2}=\frac{y-3}{4}=\frac{z-7}{5}$

$\therefore 2(2+5\lambda)+4(1-3\lambda)+5(4\lambda-1)=0$

$\Rightarrow 18\lambda+3=0\Rightarrow\lambda=-\frac{1}{6}$

Putting the value of $\lambda=-\frac{1}{6}$ in equation (i), we get

$x\left(2-\frac{5}{6}\right)+y\left(1+\frac{3}{6}\right)+z\left(\frac{-4}{6}-1\right)-\frac{9}{6}-3=0$

$\Rightarrow 7x+9y-10z-27=0$

This is the required equation of plane.

Now, the given point is $(8,-1,-19)$ and equation of parallel line is

$\frac{x}{-3}=\frac{y-5}{4}=\frac{z-2}{12}$

$\therefore$ Equation of line passing through the given point and parallel to the line is

$\frac{x-8}{-3}=\frac{y+1}{4}=\frac{z+19}{12}=k$

$\Rightarrow x=-3k+8, y=4k-1, z=12k-19$

will satisfy the plane : $7x+9y-10z-27=0$

$-21k+56+36k-9-120k+190-27=0$

$\Rightarrow -105k+210=0\Rightarrow k=2$

$\therefore x=2,\ y=7,\ z=5$

$\therefore$ Required Distance $=\sqrt{(8-2)^2+(-1-7)^2+(-19-5)^2}$

$=\sqrt{36+64+576}=\sqrt{676}=26 units$

73. We have.

$f(\theta)=\sin\theta+\int_{-\pi/2}^{\pi/2}(\sin\theta+t\cos\theta)f(t)dt$

$=\sin\theta\left\{1+\int_{-\pi/2}^{\pi/2}f(t)dt\right\}+\cos\theta\int_{-\pi/2}^{\pi/2}tf(t)dt$

$\Rightarrow f(\theta)=a\sin\theta+b\cos\theta-(i)$

where, $a=1+\int_{-\pi/2}^{\pi/2}f(t)dt$

$\Rightarrow a=1+\int_{-\pi/2}^{\pi/2}(a\sin t+b\cos t)dt-(1)$

$\Rightarrow a=1+\int_{-\pi/2}^{\pi/2}(-a\sin t+b\cos t)dt-(2)$

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Adding (1) and (2), we get

$$2a = 2 + \int_{-\pi/2}^{\pi/2} 2b\cos t\, dt$$

$$a = 1 + 2b - (ii)$$

$$\text{Also, } b = \int_{-\pi/2}^{\pi/2} tf(t)dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} t(b\cos t + a\sin t)dt - (3)$$

$$\Rightarrow b = \int_{-\pi/2}^{\pi/2} (at\sin t - tb\cos t)dt - (4)$$

Adding (3) and (4), we get

$$2b = \int_{-\pi/2}^{\pi/2} 2at\sin t dt - 4a$$

$$\Rightarrow b = 2a - (iii)$$

From (ii) and (iii), we get

$$a = -\frac{1}{3}, b = -\frac{2}{3}, \therefore f(\theta) = -\frac{1}{3}\{\sin\theta + 2\cos\theta\}$$

$$\Rightarrow \int_0^{\pi/2} f(\theta)d\theta = -\frac{1}{3}\int_0^{\pi/2} (\sin\theta + 2\cos\theta)d\theta$$

$$= -\frac{1}{3}\times 3 = -1 \Rightarrow \left|\int_0^{\frac{\pi}{2}} f(\theta)d\theta\right| = 1$$

74. $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}, A^2 = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 8 & 9 \\ 0 & -1 \end{bmatrix} \cdots$

$$A^n = \begin{bmatrix} 2^n & 2^n - (-1)^n \\ 0 & (-1)^n \end{bmatrix}$$

Also, $2A.(\text{adj}(2A)) = |2A|I$

$\Rightarrow 2A \cdot \text{adj}(2A) = 22(-2)I \Rightarrow A\text{adj}(2A) = -22I$

$$\text{Now, } |A^{10} - (\text{adj}(2A))^{10}| = \frac{1}{|A|^{10}}|A^{20} - A^{10}(\text{adj}(2A))^{10}|$$

$= \frac{1}{|A|^{10}}|A^{20} - 2^{20}I|$ - (i)

$$\because A^{20} - 220I = \begin{bmatrix} 2^{20} & 2^{20}-1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2^{20} & 0 \\ 0 & 2^{20} \end{bmatrix} = \begin{bmatrix} 0 & 2^{20}-1 \\ 0 & 1-2^{20} \end{bmatrix}$$

$\Rightarrow |A^{20} - 220I| = 0$

From (i), $|A^{10} - (\text{adj}(2A))^{10}| = 0$

Hence, $|A^4| + |A^{10} - (\text{adj}(2A))^{10}| = |A|^4 + 0 = (-2)^4 = 16$

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75. There are 98 sets of 3 consecutive integers and 97 sets of 4 consecutive integers [As all b_i's are different]

$\therefore$ Number of permutations of $b_1 b_2 b_3 b_4$ = number of permutation when $b_1 b_2 b_3$ are consecutive + number of permutations when $b_2 b_3 b_4$ are consecutive - number of permutations when $b_1 b_2 b_3 b_4$ are consecutive

$= 97 \times 98 + 97 \times 98 - 97 \left[\because b_i \neq b_j, i \neq j\right]$

$= 97 \times 195 = 18915$

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